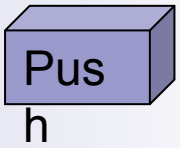
Stack

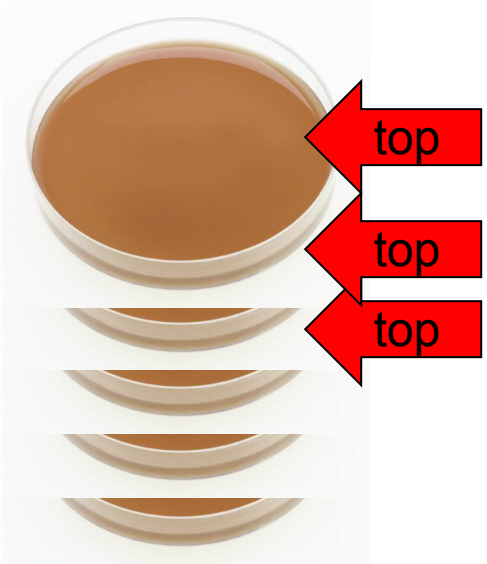


Data Structures and Algorithms



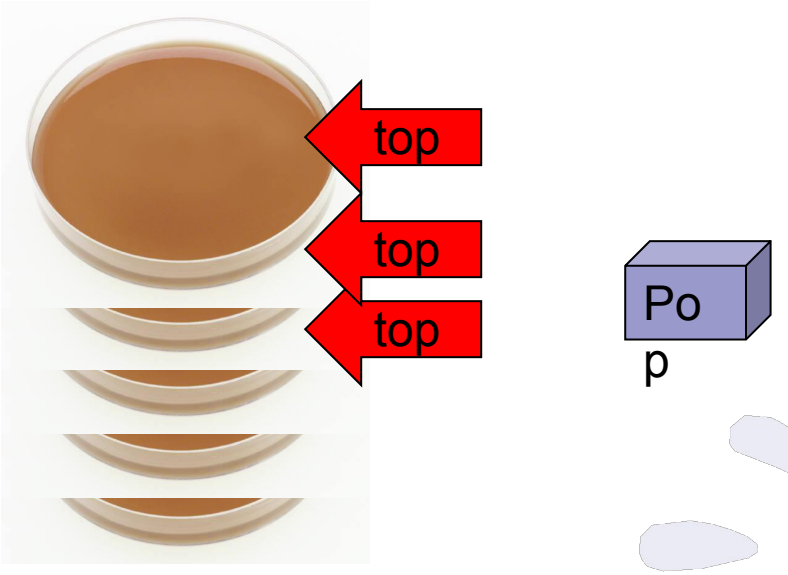
Definition





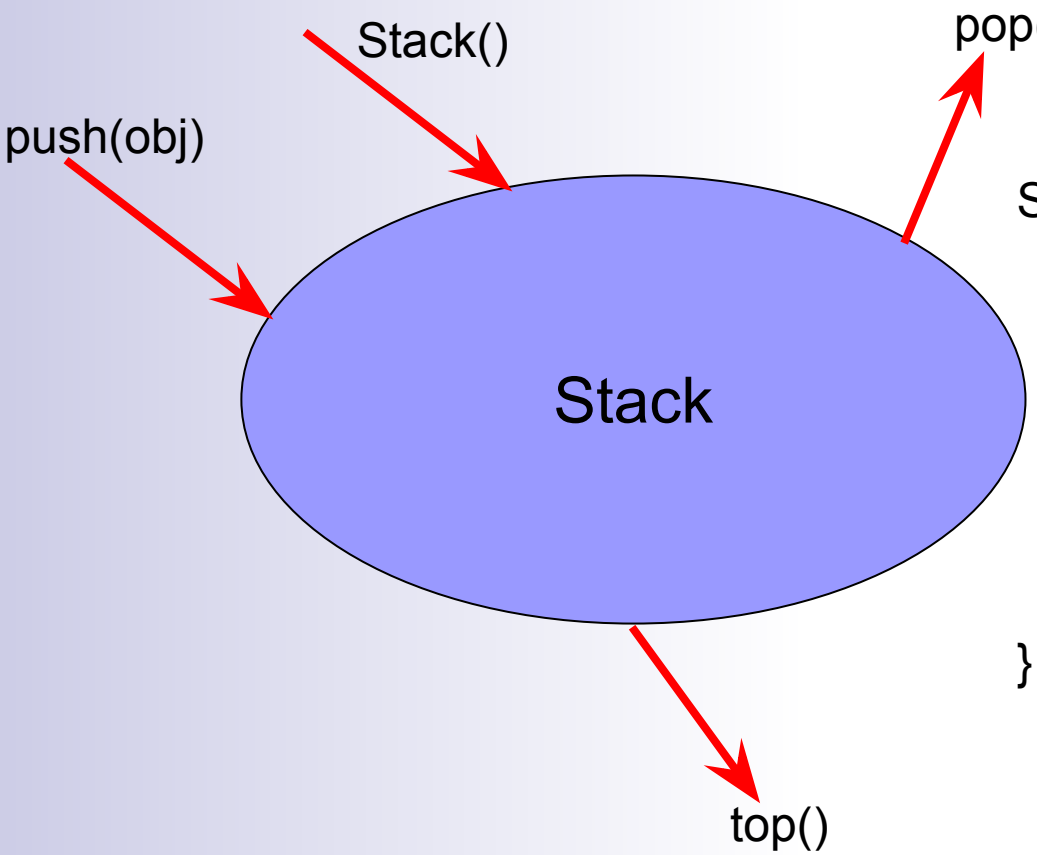


Definition



Stack is an ADT that can handle ordered collection of item in LIFO (Last In First Out)

Stack ADT

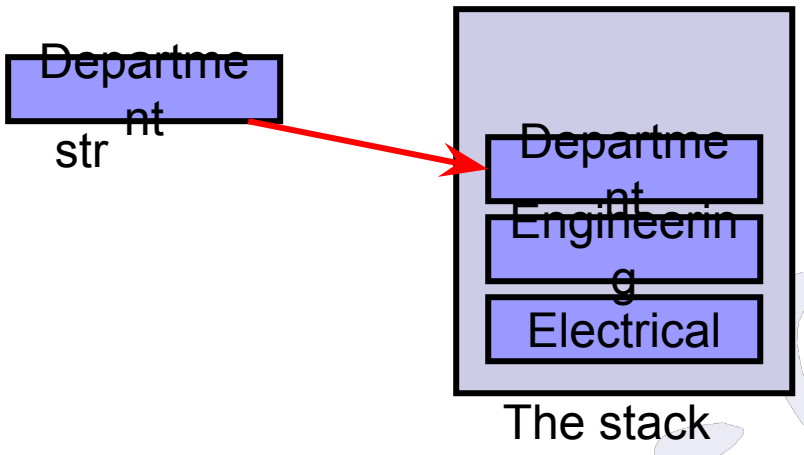



```
Stack() {
  ListNode top;

  Stack();                  // create empty stack
  void push(Object obj); // push object to stack
  Object pop();             // get & throw top element
  Object top();             // get top element
  boolean isEmpty();        // check if stack empty
}
```

Sample code

```
stack = new Stack();
stack.push('Electrical');
stack.push('Engineering');
stack.push('Department');
String str = stack.top();
stack.pop();
```



Stack Application

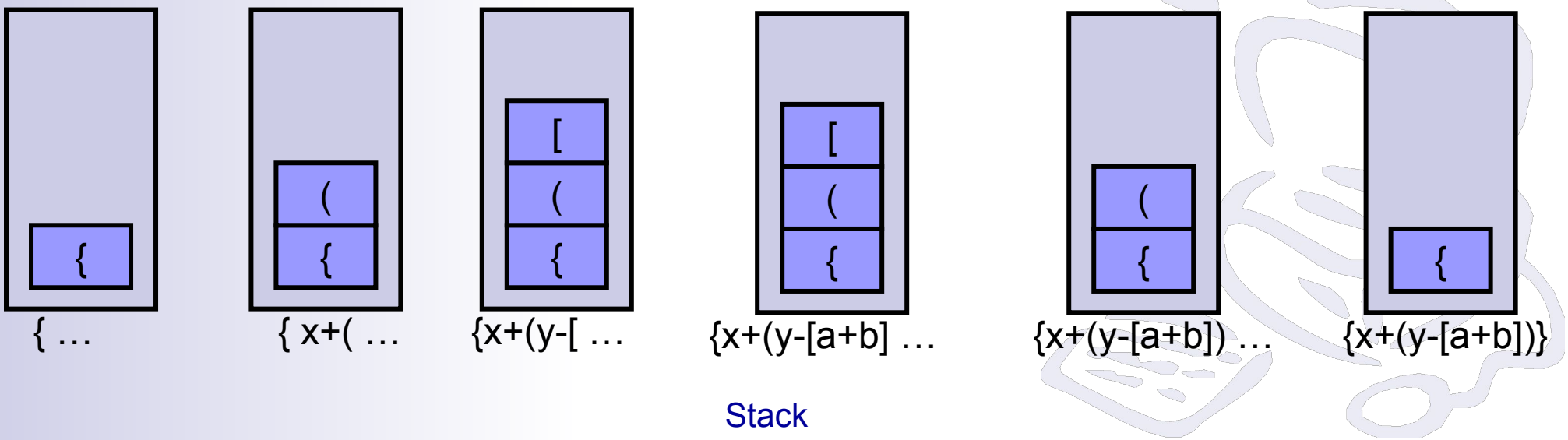
- Line editing
- Bracket matching
- Postfix calculation
- Function call stack



Matching Parenthesis

- Equal number of left and right parenthesis
- Right parenthesis is preceded by a matching left parenthesis

Expression: {x+(y-[a+b])}

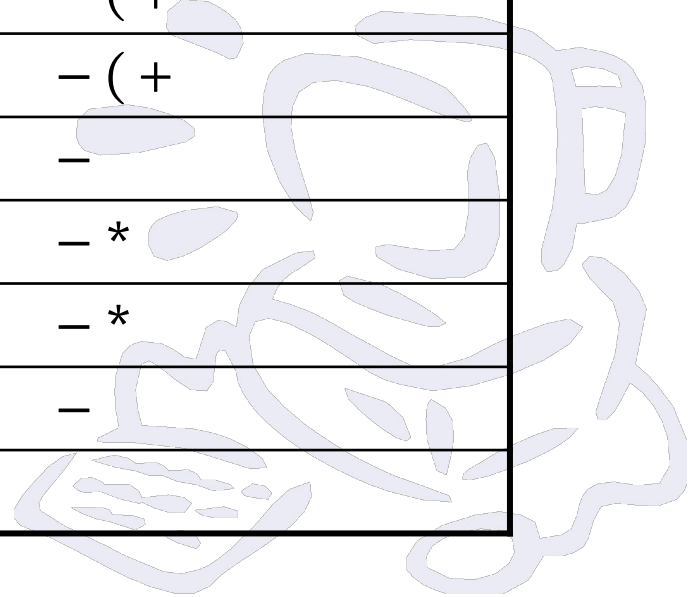


Stack

Converting Infix to Postfix Expression

Expression: 6 – (2 + 3) * 3

No	Symbol	Postfix String	Stack
1	6	6	
2	-	6	–
3	(	6	– (
4	2	6 2	– (
5	+	6 2	– (+
6	3	6 2 3	– (+
7	)	6 2 3 +	–
8	*	6 2 3 +	– *
9	3	6 2 3 + 3	– *
10		6 2 3 + 3 *	–
11		6 2 3 + 3 * –	



Converting Infix to Postfix Expression

Expression: 6 + 2 / 3 * 3

No	Symbol	Postfix String	Stack
1	6	6	
2	+	6	+
3	2	6 2	+
4	/	6 2	+ /
5	3	6 2 3	+ /
6	*	6 2 3 /	+ *
7	3	6 2 3 / 3	+ *
8		6 2 3 / 3 *	+
9		6 2 3 / 3 * +	

Expression: {6 + [(2 / 3) * 3]}

Converting Infix to Postfix Expression

Expression: 6 + 2 / 3 ^ 3

No	Symbol	Postfix String	Stack
1	6	6	
2	+	6	+
3	2	6 2	+
4	/	6 2	+ /
5	3	6 2 3	+ /
6	^	6 2 3	+ / ^
7	3	6 2 3 3	+ / ^
8		6 2 3 3 ^	+ /
9		6 2 3 3 ^ /	+
10		6 2 3 3 ^ / +	

Expression: {6 + [(2 / (3 ^ 3)]}

Stack

Evaluating a Postfix Expression

Expression: 6 2 3 + - 3 8 2 / + *

No	Symbol	Operand 1	Operand 2	Value	Stack
1	6				6
2	2				6, 2
3	3				6, 2, 3
4	+	2	3	5	6, 5
5	-	6	5	1	1
6	3	6	5	1	1, 3
7	8	6	5	1	1, 3, 8
8	2	6	5	1	1, 3, 8, 2
9	/	8	2	4	1, 3, 4
10	+	3	4	7	1, 7
11	*	1	Stack7	7	7

Assignment

Make your own code for

- Checking parenthesis
- Converting infix to postfix
- Evaluating postfix expression

